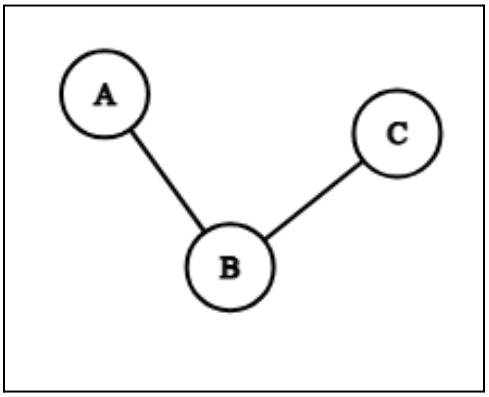
Graph Theory Introduction

1. **Node (or Vertex):** The fundamental "point" or object in a graph representing an entity (e.g., a city or a person).
2. **Edge:** The line connecting two vertices, representing a relationship or a path between them.

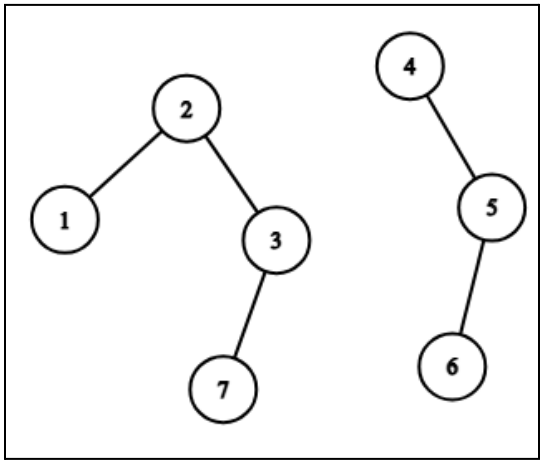
3. **Path**: A sequence of edges that allows you to travel from one specific vertex to another.

   E.g. a path from **A** to **C** is: **A** → **B** → **C**

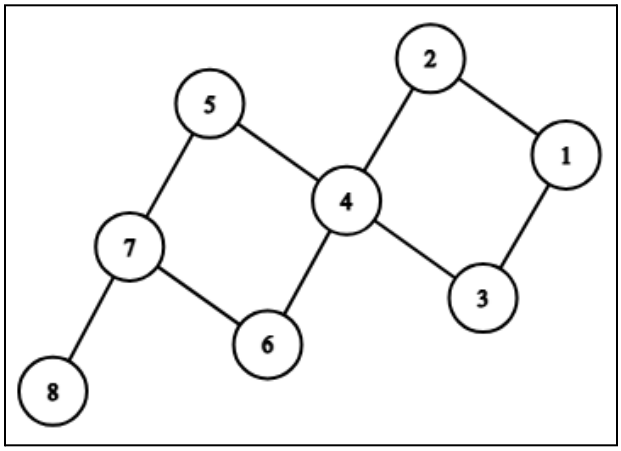


4. **Component**: A sub-group of nodes where every node is reachable from every other node in that group. If a graph has "islands" that aren't connected, each island is a component.

   For example, this graph has TWO components (or clusters)



Fork In The Road



Questions:

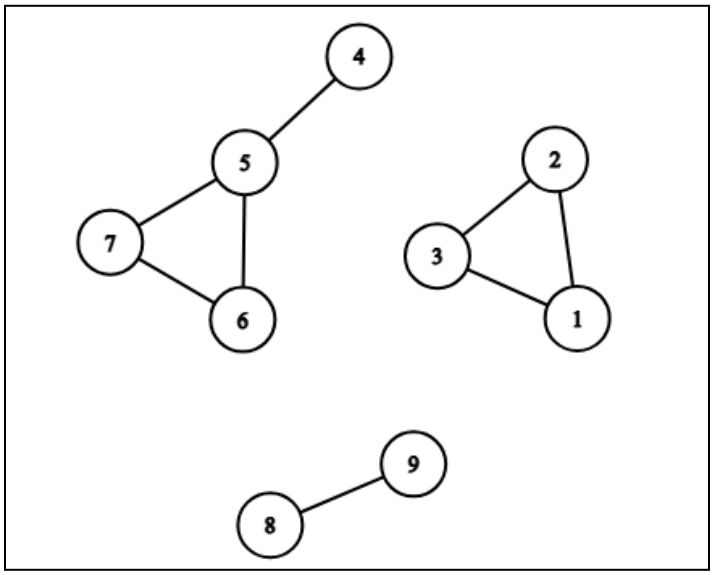
1. Is node 8 reachable from node **1**?

2. What is the minimum number of edges to get from **1** to **7**? Does that change if we add an edge connecting **2** and **5**?

3. How many ways can we reach **5** from **1**? *Assuming we can't cross the same edge twice*

4. Does this graph contain any cycles? Write them down if they exist.
   ***Note**: a graph cycle is a closed path in a graph that starts and ends at the same vertex, visiting no other vertices or edges more than once.*

Disconnected Islands



Questions:

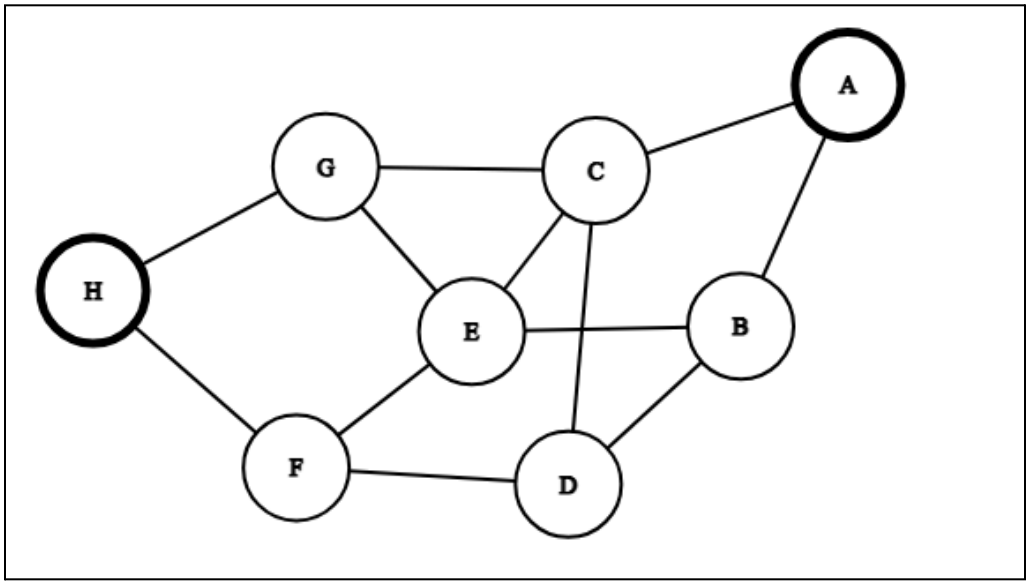
5. Can you reach node **4** starting from node **1**?
6. How many "islands" (connected components) are in this graph?

7. What is the shortest path from **4** to **7**?

   *Note: "**Shortest Path**" in this context means passing by the fewest edges*.

8. Does this graph contain any cycles?
   ***Note**: a graph cycle is a closed path in a graph that starts and ends at the same vertex, visiting no other vertices or edges more than once.*

Complex Mesh



Questions:

9. Trace two different paths from **A** to **F**. Which one is shorter?

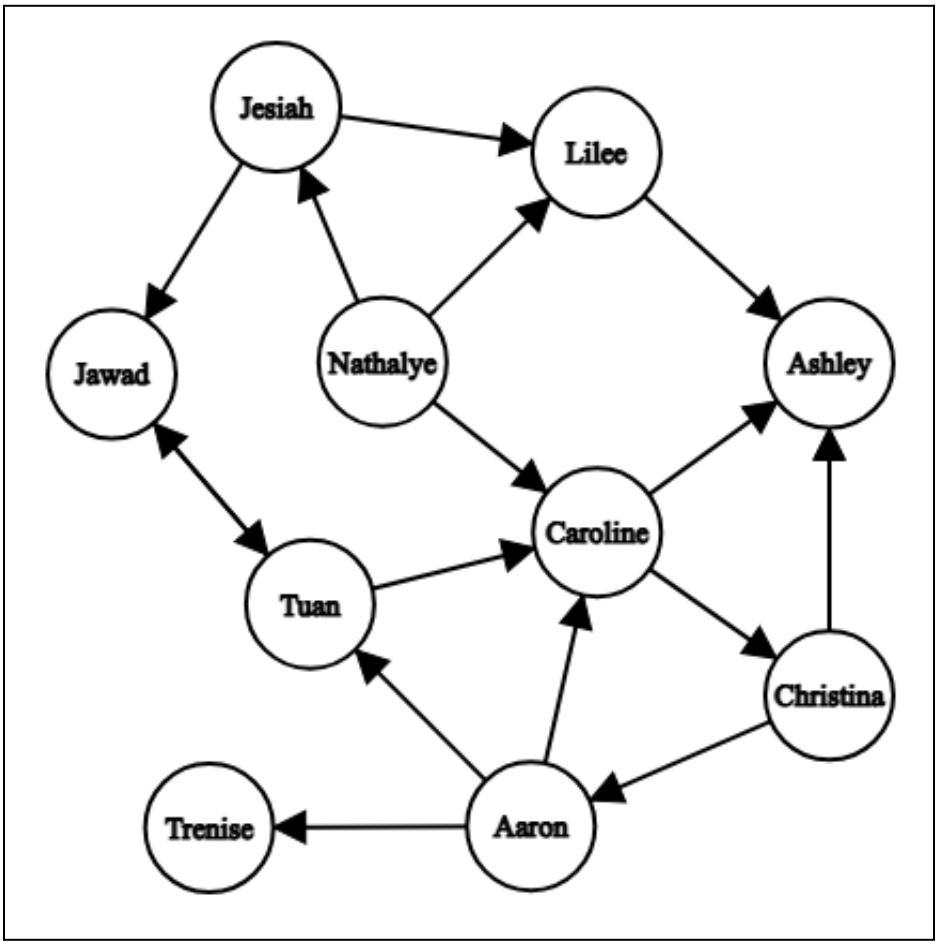
10. How many "islands" (connected components) are in this graph?

11. What is the shortest path from **A** to **H**?

12. Identify at least three cycles in this graph.

13. What is the max number of “neighbors” of a node across all nodes?

Social Media Following



Questions:

14. Which node has the highest in-degree? (highest number of followers)

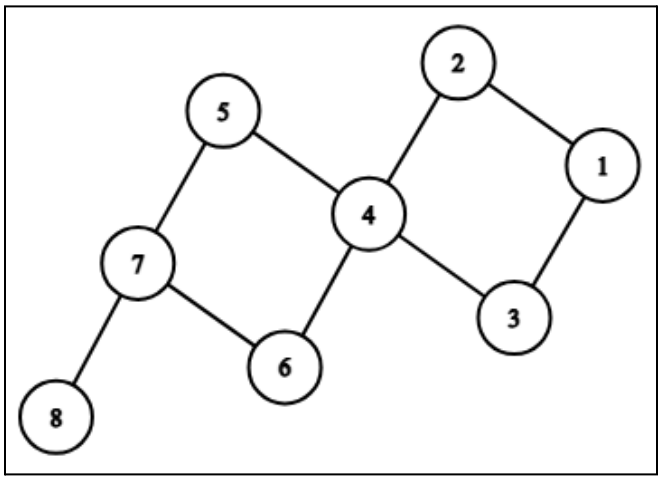
15. Which nodes have 0 out-degree? (not following anyone)

16. If we only look at "**Following**" lists, can we go from **Jesiah's** account to **Aaron's**? Can we go the opposite direction?

Graph Representation in Code

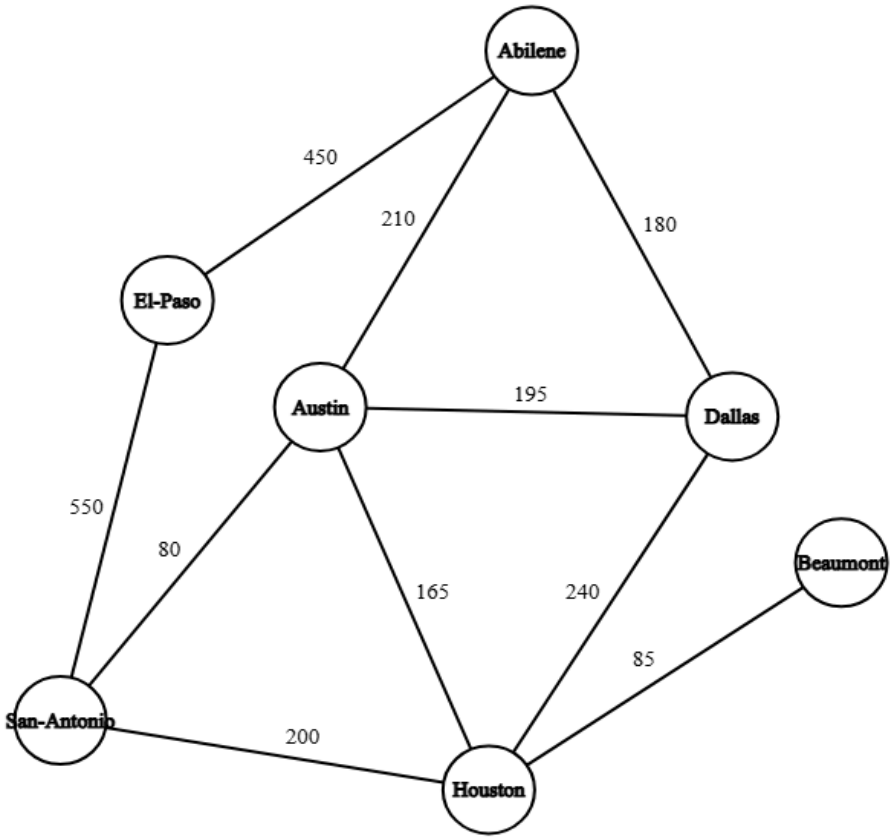
- Think of the smallest building blocks:
    - **Nodes**, and **Edges** (where each edge represents the connection Node A → Node B)
- There are two standard ways to represent this in code:

```
# Adjacency List
adj_list = {
    1: [2, 3],
    2: [4],
    3: [4],
    4: [5, 6],
    5: [7],
    6: [7],
    7: [8],
    8: []
}
# Explore neighbors of node 4
for neighbor in adj_list[4]:
    print(neighbor)
```



```
# Adjacency Matrix (8x8)
# Rows/Cols represent nodes 1-8 (index = node - 1)
matrix = [
    [0, 1, 1, 0, 0, 0, 0, 0], # Node 1
    [0, 0, 0, 1, 0, 0, 0, 0], # Node 2
    [0, 0, 0, 1, 0, 0, 0, 0], # Node 3
    [0, 0, 0, 0, 1, 1, 0, 0], # Node 4
    [0, 0, 0, 0, 0, 0, 1, 0], # Node 5
    [0, 0, 0, 0, 0, 0, 1, 0], # Node 6
    [0, 0, 0, 0, 0, 0, 0, 1], # Node 7
    [0, 0, 0, 0, 0, 0, 0, 0]  # Node 8
]
node_four_index = 3 # Explore neighbors of node 4
for index, connected in enumerate(matrix[node_four_index]):
    if connected == 1:
        print(index + 1)
```

Texas Cities



Questions:

17. What's the shortest path (sum of distances across edges) from **Houston** To **El-Paso**?

18. If the direct 200-mile route from Houston to San-Antonio is closed, what would the answer to the previous question become?

19. **Stretch example:** To optimize the state budget, the Department of Transportation must identify a "Primary Backbone" of highways. You must connect **all cities** using the **minimum total mileage** possible, ensuring no redundant loops exist.